```
import matplotlib.pyplot as plt
import numpy as np
import os

import tensorflow as tf
```

```
2024-05-04 22:05:26.655558: E
external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:9261] Unable
to register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
2024-05-04 22:05:26.655620: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:607] Unable to
register cuFFT factory: Attempting to register factory for plugin
cuFFT when one has already been registered
2024-05-04 22:05:26.702316: E
external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1515] Unable
to register cuBLAS factory: Attempting to register factory for plugin
cuBLAS when one has already been registered
2024-05-04 22:05:26.793602: I
tensorflow/core/platform/cpu_feature_guard.cc:182] This TensorFlow
binary is optimized to use available CPU instructions in performance-
critical operations.
To enable the following instructions: AVX2 FMA, in other operations,
rebuild TensorFlow with the appropriate compiler flags.
2024-05-04 22:05:28.035759: W
tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning:
Could not find TensorRT
```

Preprocessing

Read Data

```
# from google.colab import drive
# drive.mount('/content/drive')
# import os

# GOOGLE_DRIVE_PATH_AFTER_MYDRIVE = 'CS539_ANN/final_project'
# GOOGLE_DRIVE_PATH = os.path.join('drive', 'My Drive',
GOOGLE_DRIVE_PATH_AFTER_MYDRIVE)
# print(os.listdir(GOOGLE_DRIVE_PATH))

# PATH = os.path.join(GOOGLE_DRIVE_PATH,
'archive/asl_alphabet_train/asl_alphabet_train') # why is the folder
structure like this idk
BATCH_SIZE = 32
IMG_SIZE = (160, 160)

# dataset = tf.keras.utils.image_dataset_from_directory(PATH, shuffle
```

```
= True, batch_size = BATCH_SIZE, image_size = IMG_SIZE)

dataset =
tf.keras.utils.image_dataset_from_directory('archive/asl_alphabet_trai
n/asl_alphabet_train', shuffle = True, batch_size = 32, image_size =
(160, 160))

class_names = dataset.class_names
num_classes = len(class_names)
```

```
Found 87000 files belonging to 29 classes.
```

```
# Get the total number of batches in the dataset
total_batches = tf.data.experimental.cardinality(dataset).numpy()

# Calculate the number of batches for each split
val_batches = total_batches // 5  # 20% for validation
test_batches = total_batches // 5  # 20% for testing
train_batches = total_batches - val_batches - test_batches  # 60% for
training

# # Way too much data
# val_batches = 5
# test_batches = 5
# train_batches = 20

# Split the dataset into train, validation, and test datasets
train_dataset = dataset.take(train_batches)
remaining_dataset = dataset.skip(train_batches)

validation_dataset = remaining_dataset.take(val_batches)
test_dataset = remaining_dataset.skip(val_batches).take(test_batches)

print('Number of train batches: %d' %
tf.data.experimental.cardinality(train_dataset))
print('Number of validation batches: %d' %
tf.data.experimental.cardinality(validation_dataset))
print('Number of test batches: %d' %
tf.data.experimental.cardinality(test_dataset))
```

```
Number of train batches: 1633
Number of validation batches: 543
Number of test batches: 543
```

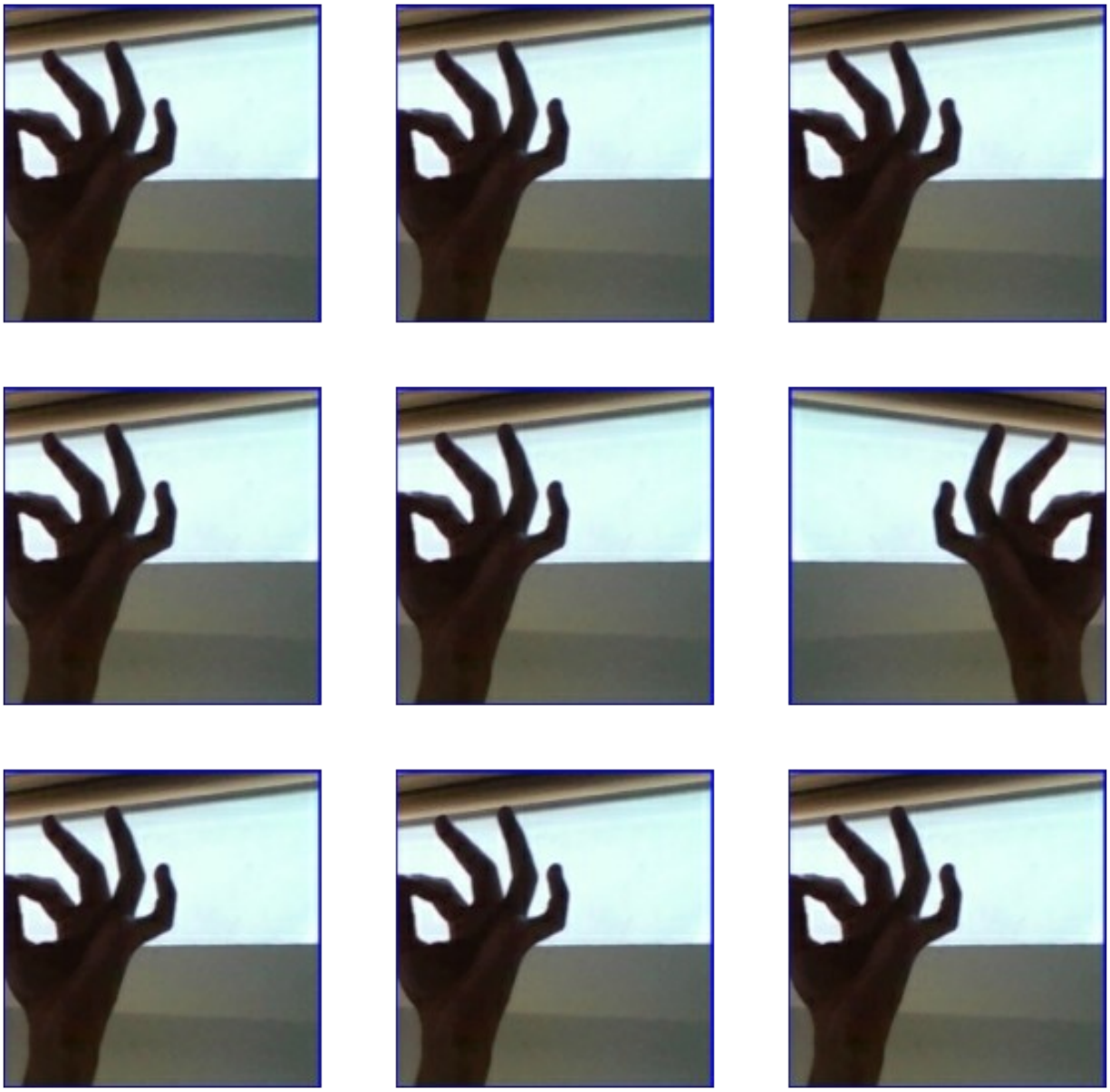
Performance Configurations

```
AUTOTUNE = tf.data.AUTOTUNE

train_dataset = train_dataset.prefetch(buffer_size=AUTOTUNE)
```

```
validation_dataset = validation_dataset.prefetch(buffer_size=AUTOTUNE)
test_dataset = test_dataset.prefetch(buffer_size=AUTOTUNE)
```

Data Augmentation and Pixel Normalization

```
data_augmentation = tf.keras.Sequential([
  tf.keras.layers.RandomFlip('horizontal'),
])

for image, _ in train_dataset.take(1):
  plt.figure(figsize=(10, 10))
  first_image = image[0]
  for i in range(9):
    ax = plt.subplot(3, 3, i + 1)
    augmented_image = data_augmentation(tf.expand_dims(first_image,
0))
    plt.imshow(augmented_image[0] / 255)
    plt.axis('off')
```



Base Model

Feature Extraction

```
preprocess_input = tf.keras.applications.efficientnet.preprocess_input


# Base Model
base_model =
tf.keras.applications.EfficientNetB0(input_shape=((IMG_SIZE) + (3,)),
                                                        include_top=False,
```

```
                                                    weights='imagenet')


image_batch, label_batch = next(iter(train_dataset))
feature_batch = base_model(image_batch)
print(feature_batch.shape)

base_model.summary()
```

```
(32, 5, 5, 1280)
Model: "efficientnetb0"
__________________________________________________________________________________________________
 Layer (type)                   Output Shape                 Param #     Connected to
==================================================================================================
 input_1 (InputLayer)           [(None, 160, 160, 3)]        0           []



 rescaling (Rescaling)          (None, 160, 160, 3)          0           ['input_1[0][0]']


 normalization (Normalizati  (None, 160, 160, 3)          7           ['rescaling[0][0]']
 on)



 rescaling_1 (Rescaling)        (None, 160, 160, 3)          0           ['normalization[0][0]']


 stem_conv_pad (ZeroPadding  (None, 161, 161, 3)          0           ['rescaling_1[0][0]']
 2D)



 stem_conv (Conv2D)             (None, 80, 80, 32)           864         ['stem_conv_pad[0][0]']


 stem_bn (BatchNormalizatio  (None, 80, 80, 32)           128         ['stem_conv[0][0]']
 n)
```

```
 stem_activation (Activatio  (None, 80, 80, 32)           0
['stem_bn[0][0]']
 n)



 block1a_dwconv (DepthwiseC  (None, 80, 80, 32)           288
['stem_activation[0][0]']
 onv2D)



 block1a_bn (BatchNormaliza  (None, 80, 80, 32)           128
['block1a_dwconv[0][0]']
 tion)



 block1a_activation (Activa  (None, 80, 80, 32)           0
['block1a_bn[0][0]']
 tion)



 block1a_se_squeeze (Global  (None, 32)                   0
['block1a_activation[0][0]']
 AveragePooling2D)



 block1a_se_reshape (Reshap  (None, 1, 1, 32)             0
['block1a_se_squeeze[0][0]']
 e)



 block1a_se_reduce (Conv2D)  (None, 1, 1, 8)              264
['block1a_se_reshape[0][0]']


 block1a_se_expand (Conv2D)  (None, 1, 1, 32)             288
['block1a_se_reduce[0][0]']


 block1a_se_excite (Multipl  (None, 80, 80, 32)           0
['block1a_activation[0][0]',
 y)
'block1a_se_expand[0][0]']


 block1a_project_conv (Conv  (None, 80, 80, 16)           512
```

```
['block1a_se_excite[0][0]']
 2D)



 block1a_project_bn (BatchN  (None, 80, 80, 16)           64
['block1a_project_conv[0][0]']
 ormalization)



 block2a_expand_conv (Conv2  (None, 80, 80, 96)           1536
['block1a_project_bn[0][0]']
 D)



 block2a_expand_bn (BatchNo  (None, 80, 80, 96)           384
['block2a_expand_conv[0][0]']
 rmalization)



 block2a_expand_activation   (None, 80, 80, 96)           0
['block2a_expand_bn[0][0]']
 (Activation)



 block2a_dwconv_pad (ZeroPa  (None, 81, 81, 96)           0
['block2a_expand_activation[0]
 dding2D)
[0]']


 block2a_dwconv (DepthwiseC  (None, 40, 40, 96)           864
['block2a_dwconv_pad[0][0]']
 onv2D)



 block2a_bn (BatchNormaliza  (None, 40, 40, 96)           384
['block2a_dwconv[0][0]']
 tion)



 block2a_activation (Activa  (None, 40, 40, 96)           0
['block2a_bn[0][0]']
 tion)
```

```
 block2a_se_squeeze (Global  (None, 96)                    0
['block2a_activation[0][0]']
 AveragePooling2D)



 block2a_se_reshape (Reshap  (None, 1, 1, 96)              0
['block2a_se_squeeze[0][0]']
 e)



 block2a_se_reduce (Conv2D)  (None, 1, 1, 4)               388
['block2a_se_reshape[0][0]']


 block2a_se_expand (Conv2D)  (None, 1, 1, 96)              480
['block2a_se_reduce[0][0]']


 block2a_se_excite (Multipl  (None, 40, 40, 96)            0
['block2a_activation[0][0]',
 y)
'block2a_se_expand[0][0]']


 block2a_project_conv (Conv  (None, 40, 40, 24)            2304
['block2a_se_excite[0][0]']
 2D)



 block2a_project_bn (BatchN  (None, 40, 40, 24)            96
['block2a_project_conv[0][0]']
 ormalization)



 block2b_expand_conv (Conv2  (None, 40, 40, 144)           3456
['block2a_project_bn[0][0]']
 D)



 block2b_expand_bn (BatchNo  (None, 40, 40, 144)           576
['block2b_expand_conv[0][0]']
 rmalization)
```

```
 block2b_expand_activation   (None, 40, 40, 144)          0
['block2b_expand_bn[0][0]']
 (Activation)



 block2b_dwconv (DepthwiseC  (None, 40, 40, 144)          1296
['block2b_expand_activation[0]
 onv2D)
[0]']


 block2b_bn (BatchNormaliza  (None, 40, 40, 144)          576
['block2b_dwconv[0][0]']
 tion)



 block2b_activation (Activa  (None, 40, 40, 144)          0
['block2b_bn[0][0]']
 tion)



 block2b_se_squeeze (Global  (None, 144)                  0
['block2b_activation[0][0]']
 AveragePooling2D)



 block2b_se_reshape (Reshap  (None, 1, 1, 144)            0
['block2b_se_squeeze[0][0]']
 e)



 block2b_se_reduce (Conv2D)  (None, 1, 1, 6)              870
['block2b_se_reshape[0][0]']


 block2b_se_expand (Conv2D)  (None, 1, 1, 144)            1008
['block2b_se_reduce[0][0]']


 block2b_se_excite (Multipl  (None, 40, 40, 144)          0
['block2b_activation[0][0]',
 y)
'block2b_se_expand[0][0]']
```

```
 block2b_project_conv (Conv  (None, 40, 40, 24)           3456
['block2b_se_excite[0][0]']
 2D)



 block2b_project_bn (BatchN  (None, 40, 40, 24)           96
['block2b_project_conv[0][0]']
 ormalization)



 block2b_drop (Dropout)      (None, 40, 40, 24)           0
['block2b_project_bn[0][0]']


 block2b_add (Add)           (None, 40, 40, 24)           0
['block2b_drop[0][0]',

'block2a_project_bn[0][0]']


 block3a_expand_conv (Conv2  (None, 40, 40, 144)          3456
['block2b_add[0][0]']
 D)



 block3a_expand_bn (BatchNo  (None, 40, 40, 144)          576
['block3a_expand_conv[0][0]']
 rmalization)



 block3a_expand_activation   (None, 40, 40, 144)          0
['block3a_expand_bn[0][0]']
 (Activation)



 block3a_dwconv_pad (ZeroPa  (None, 43, 43, 144)          0
['block3a_expand_activation[0]
 dding2D)
[0]']


 block3a_dwconv (DepthwiseC  (None, 20, 20, 144)          3600
['block3a_dwconv_pad[0][0]']
 onv2D)
```

```
 block3a_bn (BatchNormaliza  (None, 20, 20, 144)          576
['block3a_dwconv[0][0]']
 tion)



 block3a_activation (Activa  (None, 20, 20, 144)          0
['block3a_bn[0][0]']
 tion)



 block3a_se_squeeze (Global  (None, 144)                  0
['block3a_activation[0][0]']
 AveragePooling2D)



 block3a_se_reshape (Reshap  (None, 1, 1, 144)            0
['block3a_se_squeeze[0][0]']
 e)



 block3a_se_reduce (Conv2D)  (None, 1, 1, 6)              870
['block3a_se_reshape[0][0]']


 block3a_se_expand (Conv2D)  (None, 1, 1, 144)            1008
['block3a_se_reduce[0][0]']


 block3a_se_excite (Multipl  (None, 20, 20, 144)          0
['block3a_activation[0][0]',
 y)
'block3a_se_expand[0][0]']


 block3a_project_conv (Conv  (None, 20, 20, 40)           5760
['block3a_se_excite[0][0]']
 2D)



 block3a_project_bn (BatchN  (None, 20, 20, 40)           160
['block3a_project_conv[0][0]']
 ormalization)
```

```
 block3b_expand_conv (Conv2  (None, 20, 20, 240)          9600
['block3a_project_bn[0][0]']
 D)



 block3b_expand_bn (BatchNo  (None, 20, 20, 240)          960
['block3b_expand_conv[0][0]']
 rmalization)



 block3b_expand_activation   (None, 20, 20, 240)          0
['block3b_expand_bn[0][0]']
 (Activation)



 block3b_dwconv (DepthwiseC  (None, 20, 20, 240)          6000
['block3b_expand_activation[0]
 onv2D)
[0]']


 block3b_bn (BatchNormaliza  (None, 20, 20, 240)          960
['block3b_dwconv[0][0]']
 tion)



 block3b_activation (Activa  (None, 20, 20, 240)          0
['block3b_bn[0][0]']
 tion)



 block3b_se_squeeze (Global  (None, 240)                  0
['block3b_activation[0][0]']
 AveragePooling2D)



 block3b_se_reshape (Reshap  (None, 1, 1, 240)            0
['block3b_se_squeeze[0][0]']
 e)



 block3b_se_reduce (Conv2D)  (None, 1, 1, 10)             2410
['block3b_se_reshape[0][0]']
```

```
 block3b_se_expand (Conv2D)  (None, 1, 1, 240)            2640
['block3b_se_reduce[0][0]']


 block3b_se_excite (Multipl  (None, 20, 20, 240)          0
['block3b_activation[0][0]',
 y)
'block3b_se_expand[0][0]']


 block3b_project_conv (Conv  (None, 20, 20, 40)           9600
['block3b_se_excite[0][0]']
 2D)



 block3b_project_bn (BatchN  (None, 20, 20, 40)           160
['block3b_project_conv[0][0]']
 ormalization)



 block3b_drop (Dropout)      (None, 20, 20, 40)           0
['block3b_project_bn[0][0]']


 block3b_add (Add)           (None, 20, 20, 40)           0
['block3b_drop[0][0]',

'block3a_project_bn[0][0]']


 block4a_expand_conv (Conv2  (None, 20, 20, 240)          9600
['block3b_add[0][0]']
 D)



 block4a_expand_bn (BatchNo  (None, 20, 20, 240)          960
['block4a_expand_conv[0][0]']
 rmalization)



 block4a_expand_activation   (None, 20, 20, 240)          0
['block4a_expand_bn[0][0]']
 (Activation)
```

```
 block4a_dwconv_pad (ZeroPa  (None, 21, 21, 240)          0
['block4a_expand_activation[0]
 dding2D)
[0]']


 block4a_dwconv (DepthwiseC  (None, 10, 10, 240)          2160
['block4a_dwconv_pad[0][0]']
 onv2D)



 block4a_bn (BatchNormaliza  (None, 10, 10, 240)          960
['block4a_dwconv[0][0]']
 tion)



 block4a_activation (Activa  (None, 10, 10, 240)          0
['block4a_bn[0][0]']
 tion)



 block4a_se_squeeze (Global  (None, 240)                  0
['block4a_activation[0][0]']
 AveragePooling2D)



 block4a_se_reshape (Reshap  (None, 1, 1, 240)            0
['block4a_se_squeeze[0][0]']
 e)



 block4a_se_reduce (Conv2D)  (None, 1, 1, 10)             2410
['block4a_se_reshape[0][0]']


 block4a_se_expand (Conv2D)  (None, 1, 1, 240)            2640
['block4a_se_reduce[0][0]']


 block4a_se_excite (Multipl  (None, 10, 10, 240)          0
['block4a_activation[0][0]',
 y)
'block4a_se_expand[0][0]']


 block4a_project_conv (Conv  (None, 10, 10, 80)           19200
```

```
['block4a_se_excite[0][0]']
 2D)



 block4a_project_bn (BatchN  (None, 10, 10, 80)           320
['block4a_project_conv[0][0]']
 ormalization)



 block4b_expand_conv (Conv2  (None, 10, 10, 480)          38400
['block4a_project_bn[0][0]']
 D)



 block4b_expand_bn (BatchNo  (None, 10, 10, 480)          1920
['block4b_expand_conv[0][0]']
 rmalization)



 block4b_expand_activation   (None, 10, 10, 480)          0
['block4b_expand_bn[0][0]']
 (Activation)



 block4b_dwconv (DepthwiseC  (None, 10, 10, 480)          4320
['block4b_expand_activation[0]
 onv2D)
[0]']


 block4b_bn (BatchNormaliza  (None, 10, 10, 480)          1920
['block4b_dwconv[0][0]']
 tion)



 block4b_activation (Activa  (None, 10, 10, 480)          0
['block4b_bn[0][0]']
 tion)



 block4b_se_squeeze (Global  (None, 480)                  0
['block4b_activation[0][0]']
 AveragePooling2D)
```

```
 block4b_se_reshape (Reshap  (None, 1, 1, 480)            0
['block4b_se_squeeze[0][0]']
 e)



 block4b_se_reduce (Conv2D)  (None, 1, 1, 20)             9620
['block4b_se_reshape[0][0]']


 block4b_se_expand (Conv2D)  (None, 1, 1, 480)            10080
['block4b_se_reduce[0][0]']


 block4b_se_excite (Multipl  (None, 10, 10, 480)          0
['block4b_activation[0][0]',
 y)
'block4b_se_expand[0][0]']


 block4b_project_conv (Conv  (None, 10, 10, 80)           38400
['block4b_se_excite[0][0]']
 2D)



 block4b_project_bn (BatchN  (None, 10, 10, 80)           320
['block4b_project_conv[0][0]']
 ormalization)



 block4b_drop (Dropout)      (None, 10, 10, 80)           0
['block4b_project_bn[0][0]']


 block4b_add (Add)           (None, 10, 10, 80)           0
['block4b_drop[0][0]',

'block4a_project_bn[0][0]']


 block4c_expand_conv (Conv2  (None, 10, 10, 480)          38400
['block4b_add[0][0]']
 D)



 block4c_expand_bn (BatchNo  (None, 10, 10, 480)          1920
```

```
['block4c_expand_conv[0][0]']
 rmalization)



 block4c_expand_activation   (None, 10, 10, 480)          0
['block4c_expand_bn[0][0]']
 (Activation)



 block4c_dwconv (DepthwiseC  (None, 10, 10, 480)          4320
['block4c_expand_activation[0]
 onv2D)
[0]']


 block4c_bn (BatchNormaliza  (None, 10, 10, 480)          1920
['block4c_dwconv[0][0]']
 tion)



 block4c_activation (Activa  (None, 10, 10, 480)          0
['block4c_bn[0][0]']
 tion)



 block4c_se_squeeze (Global  (None, 480)                  0
['block4c_activation[0][0]']
 AveragePooling2D)



 block4c_se_reshape (Reshap  (None, 1, 1, 480)            0
['block4c_se_squeeze[0][0]']
 e)



 block4c_se_reduce (Conv2D)  (None, 1, 1, 20)             9620
['block4c_se_reshape[0][0]']


 block4c_se_expand (Conv2D)  (None, 1, 1, 480)            10080
['block4c_se_reduce[0][0]']


 block4c_se_excite (Multipl  (None, 10, 10, 480)          0
['block4c_activation[0][0]',
```

```
 y)
'block4c_se_expand[0][0]']


 block4c_project_conv (Conv  (None, 10, 10, 80)           38400
['block4c_se_excite[0][0]']
 2D)



 block4c_project_bn (BatchN  (None, 10, 10, 80)           320
['block4c_project_conv[0][0]']
 ormalization)



 block4c_drop (Dropout)      (None, 10, 10, 80)           0
['block4c_project_bn[0][0]']


 block4c_add (Add)           (None, 10, 10, 80)           0
['block4c_drop[0][0]',

'block4b_add[0][0]']


 block5a_expand_conv (Conv2  (None, 10, 10, 480)          38400
['block4c_add[0][0]']
 D)



 block5a_expand_bn (BatchNo  (None, 10, 10, 480)          1920
['block5a_expand_conv[0][0]']
 rmalization)



 block5a_expand_activation   (None, 10, 10, 480)          0
['block5a_expand_bn[0][0]']
 (Activation)



 block5a_dwconv (DepthwiseC  (None, 10, 10, 480)          12000
['block5a_expand_activation[0]
 onv2D)
[0]']


 block5a_bn (BatchNormaliza  (None, 10, 10, 480)          1920
```

```
['block5a_dwconv[0][0]']
 tion)



 block5a_activation (Activa  (None, 10, 10, 480)          0
['block5a_bn[0][0]']
 tion)



 block5a_se_squeeze (Global  (None, 480)                  0
['block5a_activation[0][0]']
 AveragePooling2D)



 block5a_se_reshape (Reshap  (None, 1, 1, 480)            0
['block5a_se_squeeze[0][0]']
 e)



 block5a_se_reduce (Conv2D)  (None, 1, 1, 20)             9620
['block5a_se_reshape[0][0]']


 block5a_se_expand (Conv2D)  (None, 1, 1, 480)            10080
['block5a_se_reduce[0][0]']


 block5a_se_excite (Multipl  (None, 10, 10, 480)          0
['block5a_activation[0][0]',
 y)
'block5a_se_expand[0][0]']


 block5a_project_conv (Conv  (None, 10, 10, 112)          53760
['block5a_se_excite[0][0]']
 2D)



 block5a_project_bn (BatchN  (None, 10, 10, 112)          448
['block5a_project_conv[0][0]']
 ormalization)



 block5b_expand_conv (Conv2  (None, 10, 10, 672)          75264
['block5a_project_bn[0][0]']
```

```
 D)



 block5b_expand_bn (BatchNo  (None, 10, 10, 672)          2688
['block5b_expand_conv[0][0]']
 rmalization)



 block5b_expand_activation   (None, 10, 10, 672)          0
['block5b_expand_bn[0][0]']
 (Activation)



 block5b_dwconv (DepthwiseC  (None, 10, 10, 672)          16800
['block5b_expand_activation[0]
 onv2D)
[0]']


 block5b_bn (BatchNormaliza  (None, 10, 10, 672)          2688
['block5b_dwconv[0][0]']
 tion)



 block5b_activation (Activa  (None, 10, 10, 672)          0
['block5b_bn[0][0]']
 tion)



 block5b_se_squeeze (Global  (None, 672)                  0
['block5b_activation[0][0]']
 AveragePooling2D)



 block5b_se_reshape (Reshap  (None, 1, 1, 672)            0
['block5b_se_squeeze[0][0]']
 e)



 block5b_se_reduce (Conv2D)  (None, 1, 1, 28)             18844
['block5b_se_reshape[0][0]']


 block5b_se_expand (Conv2D)  (None, 1, 1, 672)            19488
```

```
['block5b_se_reduce[0][0]']


 block5b_se_excite (Multipl  (None, 10, 10, 672)          0
['block5b_activation[0][0]',
 y)
'block5b_se_expand[0][0]']


 block5b_project_conv (Conv  (None, 10, 10, 112)          75264
['block5b_se_excite[0][0]']
 2D)



 block5b_project_bn (BatchN  (None, 10, 10, 112)          448
['block5b_project_conv[0][0]']
 ormalization)



 block5b_drop (Dropout)      (None, 10, 10, 112)          0
['block5b_project_bn[0][0]']


 block5b_add (Add)           (None, 10, 10, 112)          0
['block5b_drop[0][0]',

'block5a_project_bn[0][0]']


 block5c_expand_conv (Conv2  (None, 10, 10, 672)          75264
['block5b_add[0][0]']
 D)



 block5c_expand_bn (BatchNo  (None, 10, 10, 672)          2688
['block5c_expand_conv[0][0]']
 rmalization)



 block5c_expand_activation   (None, 10, 10, 672)          0
['block5c_expand_bn[0][0]']
 (Activation)



 block5c_dwconv (DepthwiseC  (None, 10, 10, 672)          16800
['block5c_expand_activation[0]
```

```
 onv2D)
[0]']


 block5c_bn (BatchNormaliza  (None, 10, 10, 672)          2688
['block5c_dwconv[0][0]']
 tion)



 block5c_activation (Activa  (None, 10, 10, 672)          0
['block5c_bn[0][0]']
 tion)



 block5c_se_squeeze (Global  (None, 672)                  0
['block5c_activation[0][0]']
 AveragePooling2D)



 block5c_se_reshape (Reshap  (None, 1, 1, 672)            0
['block5c_se_squeeze[0][0]']
 e)



 block5c_se_reduce (Conv2D)  (None, 1, 1, 28)             18844
['block5c_se_reshape[0][0]']


 block5c_se_expand (Conv2D)  (None, 1, 1, 672)            19488
['block5c_se_reduce[0][0]']


 block5c_se_excite (Multipl  (None, 10, 10, 672)          0
['block5c_activation[0][0]',
 y)
'block5c_se_expand[0][0]']


 block5c_project_conv (Conv  (None, 10, 10, 112)          75264
['block5c_se_excite[0][0]']
 2D)



 block5c_project_bn (BatchN  (None, 10, 10, 112)          448
['block5c_project_conv[0][0]']
 ormalization)
```

```
 block5c_drop (Dropout)      (None, 10, 10, 112)          0
['block5c_project_bn[0][0]']


 block5c_add (Add)           (None, 10, 10, 112)          0
['block5c_drop[0][0]',

'block5b_add[0][0]']


 block6a_expand_conv (Conv2  (None, 10, 10, 672)          75264
['block5c_add[0][0]']
 D)



 block6a_expand_bn (BatchNo  (None, 10, 10, 672)          2688
['block6a_expand_conv[0][0]']
 rmalization)



 block6a_expand_activation   (None, 10, 10, 672)          0
['block6a_expand_bn[0][0]']
 (Activation)



 block6a_dwconv_pad (ZeroPa  (None, 13, 13, 672)          0
['block6a_expand_activation[0]
 dding2D)
[0]']


 block6a_dwconv (DepthwiseC  (None, 5, 5, 672)            16800
['block6a_dwconv_pad[0][0]']
 onv2D)



 block6a_bn (BatchNormaliza  (None, 5, 5, 672)            2688
['block6a_dwconv[0][0]']
 tion)



 block6a_activation (Activa  (None, 5, 5, 672)            0
['block6a_bn[0][0]']
```

```
 tion)



 block6a_se_squeeze (Global  (None, 672)                  0
['block6a_activation[0][0]']
 AveragePooling2D)



 block6a_se_reshape (Reshap  (None, 1, 1, 672)            0
['block6a_se_squeeze[0][0]']
 e)



 block6a_se_reduce (Conv2D)  (None, 1, 1, 28)             18844
['block6a_se_reshape[0][0]']


 block6a_se_expand (Conv2D)  (None, 1, 1, 672)            19488
['block6a_se_reduce[0][0]']


 block6a_se_excite (Multipl  (None, 5, 5, 672)            0
['block6a_activation[0][0]',
 y)
'block6a_se_expand[0][0]']


 block6a_project_conv (Conv  (None, 5, 5, 192)            129024
['block6a_se_excite[0][0]']
 2D)



 block6a_project_bn (BatchN  (None, 5, 5, 192)            768
['block6a_project_conv[0][0]']
 ormalization)



 block6b_expand_conv (Conv2  (None, 5, 5, 1152)           221184
['block6a_project_bn[0][0]']
 D)



 block6b_expand_bn (BatchNo  (None, 5, 5, 1152)           4608
['block6b_expand_conv[0][0]']
 rmalization)
```

```
 block6b_expand_activation   (None, 5, 5, 1152)           0
['block6b_expand_bn[0][0]']
 (Activation)



 block6b_dwconv (DepthwiseC  (None, 5, 5, 1152)           28800
['block6b_expand_activation[0]
 onv2D)
[0]']


 block6b_bn (BatchNormaliza  (None, 5, 5, 1152)           4608
['block6b_dwconv[0][0]']
 tion)



 block6b_activation (Activa  (None, 5, 5, 1152)           0
['block6b_bn[0][0]']
 tion)



 block6b_se_squeeze (Global  (None, 1152)                 0
['block6b_activation[0][0]']
 AveragePooling2D)



 block6b_se_reshape (Reshap  (None, 1, 1, 1152)           0
['block6b_se_squeeze[0][0]']
 e)



 block6b_se_reduce (Conv2D)  (None, 1, 1, 48)             55344
['block6b_se_reshape[0][0]']


 block6b_se_expand (Conv2D)  (None, 1, 1, 1152)           56448
['block6b_se_reduce[0][0]']


 block6b_se_excite (Multipl  (None, 5, 5, 1152)           0
['block6b_activation[0][0]',
 y)
'block6b_se_expand[0][0]']
```

```
 block6b_project_conv (Conv  (None, 5, 5, 192)            221184
['block6b_se_excite[0][0]']
 2D)



 block6b_project_bn (BatchN  (None, 5, 5, 192)            768
['block6b_project_conv[0][0]']
 ormalization)



 block6b_drop (Dropout)      (None, 5, 5, 192)            0
['block6b_project_bn[0][0]']


 block6b_add (Add)           (None, 5, 5, 192)            0
['block6b_drop[0][0]',

'block6a_project_bn[0][0]']


 block6c_expand_conv (Conv2  (None, 5, 5, 1152)           221184
['block6b_add[0][0]']
 D)



 block6c_expand_bn (BatchNo  (None, 5, 5, 1152)           4608
['block6c_expand_conv[0][0]']
 rmalization)



 block6c_expand_activation   (None, 5, 5, 1152)           0
['block6c_expand_bn[0][0]']
 (Activation)



 block6c_dwconv (DepthwiseC  (None, 5, 5, 1152)           28800
['block6c_expand_activation[0]
 onv2D)
[0]']


 block6c_bn (BatchNormaliza  (None, 5, 5, 1152)           4608
['block6c_dwconv[0][0]']
 tion)
```

```
 block6c_activation (Activa  (None, 5, 5, 1152)           0
['block6c_bn[0][0]']
 tion)



 block6c_se_squeeze (Global  (None, 1152)                 0
['block6c_activation[0][0]']
 AveragePooling2D)



 block6c_se_reshape (Reshap  (None, 1, 1, 1152)           0
['block6c_se_squeeze[0][0]']
 e)



 block6c_se_reduce (Conv2D)  (None, 1, 1, 48)             55344
['block6c_se_reshape[0][0]']


 block6c_se_expand (Conv2D)  (None, 1, 1, 1152)           56448
['block6c_se_reduce[0][0]']


 block6c_se_excite (Multipl  (None, 5, 5, 1152)           0
['block6c_activation[0][0]',
 y)
'block6c_se_expand[0][0]']


 block6c_project_conv (Conv  (None, 5, 5, 192)            221184
['block6c_se_excite[0][0]']
 2D)



 block6c_project_bn (BatchN  (None, 5, 5, 192)            768
['block6c_project_conv[0][0]']
 ormalization)



 block6c_drop (Dropout)      (None, 5, 5, 192)            0
['block6c_project_bn[0][0]']
```

```
 block6c_add (Add)           (None, 5, 5, 192)            0
['block6c_drop[0][0]',

'block6b_add[0][0]']


 block6d_expand_conv (Conv2  (None, 5, 5, 1152)           221184
['block6c_add[0][0]']
 D)



 block6d_expand_bn (BatchNo  (None, 5, 5, 1152)           4608
['block6d_expand_conv[0][0]']
 rmalization)



 block6d_expand_activation   (None, 5, 5, 1152)           0
['block6d_expand_bn[0][0]']
 (Activation)



 block6d_dwconv (DepthwiseC  (None, 5, 5, 1152)           28800
['block6d_expand_activation[0]
 onv2D)
[0]']


 block6d_bn (BatchNormaliza  (None, 5, 5, 1152)           4608
['block6d_dwconv[0][0]']
 tion)



 block6d_activation (Activa  (None, 5, 5, 1152)           0
['block6d_bn[0][0]']
 tion)



 block6d_se_squeeze (Global  (None, 1152)                 0
['block6d_activation[0][0]']
 AveragePooling2D)



 block6d_se_reshape (Reshap  (None, 1, 1, 1152)           0
['block6d_se_squeeze[0][0]']
 e)
```

```
 block6d_se_reduce (Conv2D)  (None, 1, 1, 48)             55344
['block6d_se_reshape[0][0]']


 block6d_se_expand (Conv2D)  (None, 1, 1, 1152)           56448
['block6d_se_reduce[0][0]']


 block6d_se_excite (Multipl  (None, 5, 5, 1152)           0
['block6d_activation[0][0]',
 y)
'block6d_se_expand[0][0]']


 block6d_project_conv (Conv  (None, 5, 5, 192)            221184
['block6d_se_excite[0][0]']
 2D)



 block6d_project_bn (BatchN  (None, 5, 5, 192)            768
['block6d_project_conv[0][0]']
 ormalization)



 block6d_drop (Dropout)      (None, 5, 5, 192)            0
['block6d_project_bn[0][0]']


 block6d_add (Add)           (None, 5, 5, 192)            0
['block6d_drop[0][0]',

'block6c_add[0][0]']


 block7a_expand_conv (Conv2  (None, 5, 5, 1152)           221184
['block6d_add[0][0]']
 D)



 block7a_expand_bn (BatchNo  (None, 5, 5, 1152)           4608
['block7a_expand_conv[0][0]']
 rmalization)
```

```
 block7a_expand_activation   (None, 5, 5, 1152)           0
['block7a_expand_bn[0][0]']
 (Activation)



 block7a_dwconv (DepthwiseC  (None, 5, 5, 1152)           10368
['block7a_expand_activation[0]
 onv2D)
[0]']


 block7a_bn (BatchNormaliza  (None, 5, 5, 1152)           4608
['block7a_dwconv[0][0]']
 tion)



 block7a_activation (Activa  (None, 5, 5, 1152)           0
['block7a_bn[0][0]']
 tion)



 block7a_se_squeeze (Global  (None, 1152)                 0
['block7a_activation[0][0]']
 AveragePooling2D)



 block7a_se_reshape (Reshap  (None, 1, 1, 1152)           0
['block7a_se_squeeze[0][0]']
 e)



 block7a_se_reduce (Conv2D)  (None, 1, 1, 48)             55344
['block7a_se_reshape[0][0]']


 block7a_se_expand (Conv2D)  (None, 1, 1, 1152)           56448
['block7a_se_reduce[0][0]']


 block7a_se_excite (Multipl  (None, 5, 5, 1152)           0
['block7a_activation[0][0]',
 y)
'block7a_se_expand[0][0]']


 block7a_project_conv (Conv  (None, 5, 5, 320)            368640
```

```
['block7a_se_excite[0][0]']
 2D)



 block7a_project_bn (BatchN  (None, 5, 5, 320)            1280
['block7a_project_conv[0][0]']
 ormalization)



 top_conv (Conv2D)           (None, 5, 5, 1280)           409600
['block7a_project_bn[0][0]']


 top_bn (BatchNormalization  (None, 5, 5, 1280)           5120
['top_conv[0][0]']
 )



 top_activation (Activation  (None, 5, 5, 1280)           0
['top_bn[0][0]']
 )



==================================================================================================
Total params: 4049571 (15.45 MB)
Trainable params: 4007548 (15.29 MB)
Non-trainable params: 42023 (164.16 KB)
__________________________________________________________________________________________________
```

Freeze Base Model

```
base_model.trainable = False
```

Add Classification Head

```
global_average_layer = tf.keras.layers.GlobalAveragePooling2D()
feature_batch_average = global_average_layer(feature_batch)

prediction_layer = tf.keras.layers.Dense(num_classes,
activation="sigmoid")

prediction_batch = prediction_layer(feature_batch_average)

inputs = tf.keras.Input(shape=(160, 160, 3))
```

```
x = data_augmentation(inputs)

x = preprocess_input(x)
x = base_model(x, training=False)
x = global_average_layer(x)

outputs = prediction_layer(x)
model = tf.keras.Model(inputs, outputs)

model.summary()
```

```
Model: "model"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_2 (InputLayer)        [(None, 160, 160, 3)]     0

 sequential (Sequential)     (None, 160, 160, 3)       0

 efficientnetb0 (Functional  (None, 5, 5, 1280)        4049571
 )

 global_average_pooling2d (  (None, 1280)              0
 GlobalAveragePooling2D)

 dense (Dense)               (None, 29)                37149

=================================================================
Total params: 4086720 (15.59 MB)
Trainable params: 37149 (145.11 KB)
Non-trainable params: 4049571 (15.45 MB)
_________________________________________________________________
```

Compile Model

```
base_learning_rate = 0.00001
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=base_le
arning_rate),

loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=False),
              metrics=['accuracy'])
```

Train Model

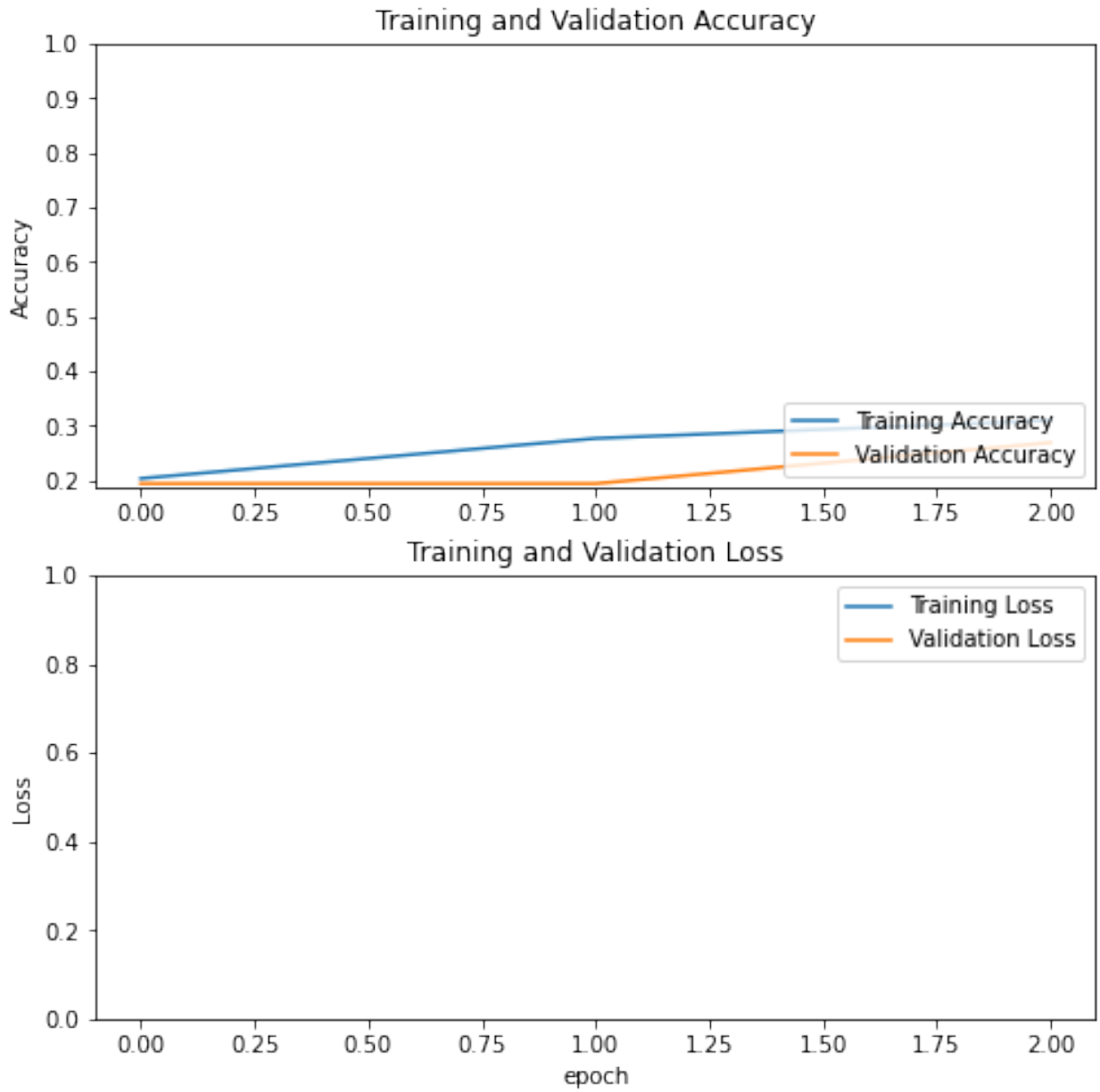
```
initial_epochs = 3

loss0, accuracy0 = model.evaluate(validation_dataset)
history = model.fit(train_dataset,
```

```
                    epochs=initial_epochs,
                    validation_data=validation_dataset)
```

```
5/5 [==============================] - 2s 275ms/step - loss: 3.1052 -
accuracy: 0.1562
Epoch 1/3
20/20 [==============================] - 7s 350ms/step - loss: 3.0248
- accuracy: 0.2031 - val_loss: 3.0624 - val_accuracy: 0.1937
Epoch 2/3
20/20 [==============================] - 6s 302ms/step - loss: 2.9130
- accuracy: 0.2766 - val_loss: 3.0241 - val_accuracy: 0.1937
Epoch 3/3
20/20 [==============================] - 6s 308ms/step - loss: 2.8368
- accuracy: 0.3094 - val_loss: 2.9370 - val_accuracy: 0.2688
```

Performance Metrics

```
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']

loss = history.history['loss']
val_loss = history.history['val_loss']

plt.figure(figsize=(8, 8))
plt.subplot(2, 1, 1)
plt.plot(acc, label='Training Accuracy')
plt.plot(val_acc, label='Validation Accuracy')
plt.legend(loc='lower right')
plt.ylabel('Accuracy')
plt.ylim([min(plt.ylim()),1])
plt.title('Training and Validation Accuracy')

plt.subplot(2, 1, 2)
plt.plot(loss, label='Training Loss')
plt.plot(val_loss, label='Validation Loss')
plt.legend(loc='upper right')
plt.ylabel('Loss')
plt.ylim([0,1.0])
plt.title('Training and Validation Loss')
plt.xlabel('epoch')
plt.show()
```



Fine Tune

Unfreeze Base Model

```
start = len(base_model.layers) - 3
print(start)

for layer in model.layers[:start]:
   layer.trainable = False
for layer in model.layers[start:]:
   layer.trainable = True
```

```
235
```

Recompile Model

```
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=base_le
arning_rate/100),

loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=False),
              metrics=['accuracy'])
```

Fine Tune Training

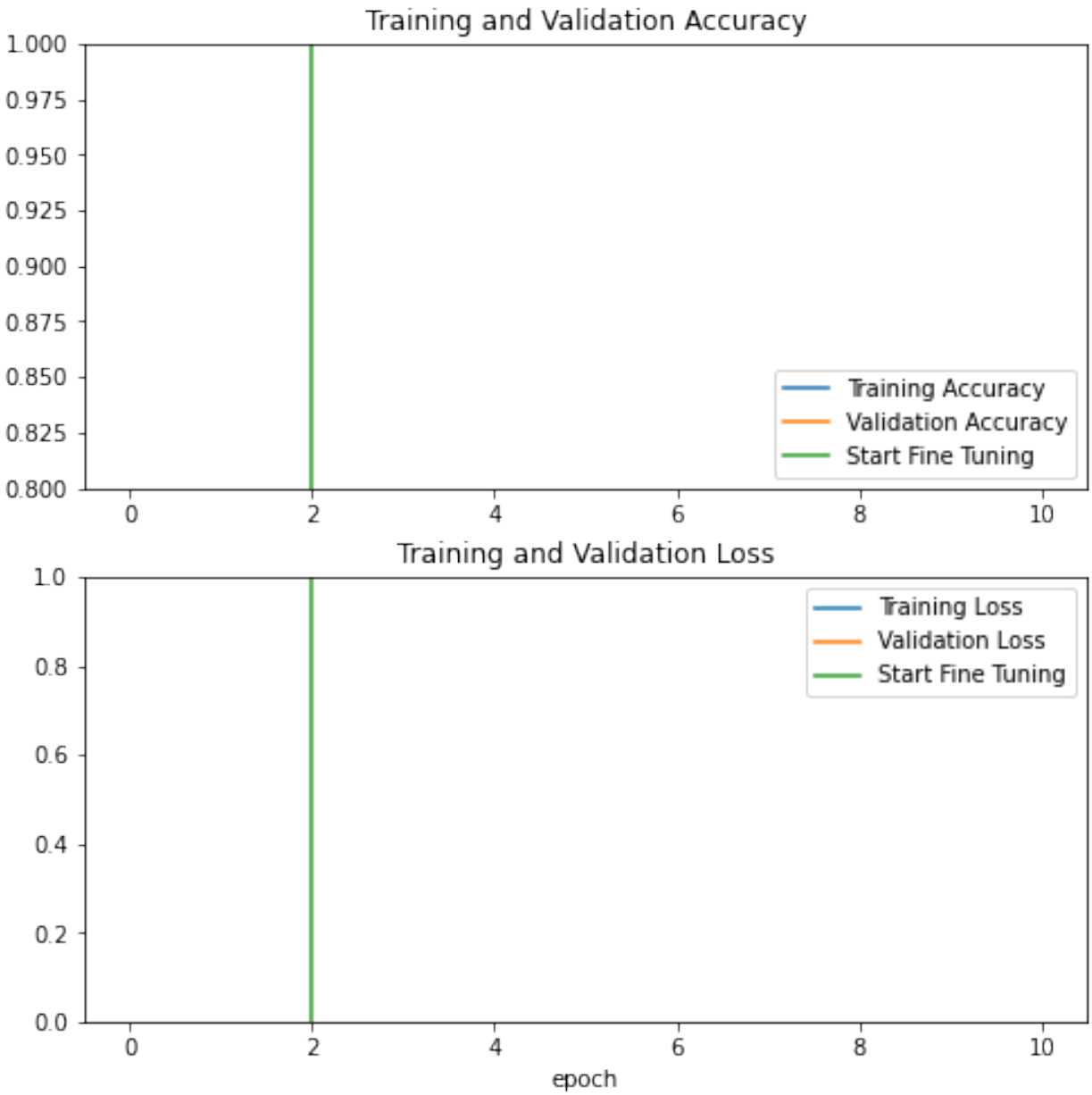
```
fine_tune_epochs = 3
total_epochs =  initial_epochs + fine_tune_epochs

history_fine = model.fit(train_dataset,
                         epochs=total_epochs,
                         initial_epoch=history.epoch[-1],
                         validation_data=validation_dataset)

acc += history_fine.history['accuracy']
val_acc += history_fine.history['val_accuracy']

loss += history_fine.history['loss']
val_loss += history_fine.history['val_loss']
```

```
Epoch 3/6

20/20 [==============================] - 11s 373ms/step - loss: 2.7961
- accuracy: 0.3359 - val_loss: 2.9130 - val_accuracy: 0.2562
Epoch 4/6
20/20 [==============================] - 7s 332ms/step - loss: 2.8012
- accuracy: 0.3234 - val_loss: 2.8817 - val_accuracy: 0.2875
Epoch 5/6
20/20 [==============================] - 7s 347ms/step - loss: 2.7682
- accuracy: 0.3656 - val_loss: 2.9139 - val_accuracy: 0.2937
Epoch 6/6
20/20 [==============================] - 7s 326ms/step - loss: 2.7773
- accuracy: 0.3359 - val_loss: 2.8851 - val_accuracy: 0.3063
```

Performance Metrics

```
acc += history_fine.history['accuracy']
val_acc += history_fine.history['val_accuracy']

loss += history_fine.history['loss']
val_loss += history_fine.history['val_loss']

plt.figure(figsize=(8, 8))
```

```
plt.subplot(2, 1, 1)
plt.plot(acc, label='Training Accuracy')
plt.plot(val_acc, label='Validation Accuracy')
plt.ylim([0.8, 1])
plt.plot([initial_epochs-1,initial_epochs-1],
          plt.ylim(), label='Start Fine Tuning')
plt.legend(loc='lower right')
plt.title('Training and Validation Accuracy')

plt.subplot(2, 1, 2)
plt.plot(loss, label='Training Loss')
plt.plot(val_loss, label='Validation Loss')
plt.ylim([0, 1.0])
plt.plot([initial_epochs-1,initial_epochs-1],
         plt.ylim(), label='Start Fine Tuning')
plt.legend(loc='upper right')
plt.title('Training and Validation Loss')
plt.xlabel('epoch')
plt.show()

loss, accuracy = model.evaluate(test_dataset)
print('Test accuracy :', accuracy)
```




```
5/5 [==============================] - 2s 222ms/step - loss: 2.8688 -
accuracy: 0.2812
Test accuracy : 0.28125

# model.save('EfficientNetB0_4-27_940PM.keras')

loaded_model = tf.keras.saving.load_model('EfficientNetB0_4-
27_940PM.keras')
model = loaded_model

from sklearn.metrics import confusion_matrix
import numpy as np
import matplotlib.pyplot as plt
```

```python
import itertools

# Get the predictions for the test dataset
y_pred = []
y_true = []

for x, y in test_dataset:
    y_pred.extend(model.predict(x))
    y_true.extend(y.numpy())

# Convert the predictions and true labels to NumPy arrays
y_pred = np.array(y_pred)
y_true = np.array(y_true)

# Convert the predictions to class labels
y_pred_classes = np.argmax(y_pred, axis=1)

# Generate the confusion matrix
confusion_mtx = confusion_matrix(y_true, y_pred_classes)
```

```
1/1 [==============================] - 0s 228ms/step
1/1 [==============================] - 0s 218ms/step
1/1 [==============================] - 0s 222ms/step
1/1 [==============================] - 0s 222ms/step
1/1 [==============================] - 0s 212ms/step
1/1 [==============================] - 0s 236ms/step
1/1 [==============================] - 0s 217ms/step
1/1 [==============================] - 0s 211ms/step
1/1 [==============================] - 0s 243ms/step
1/1 [==============================] - 0s 239ms/step
1/1 [==============================] - 0s 224ms/step
1/1 [==============================] - 0s 226ms/step
1/1 [==============================] - 0s 225ms/step
1/1 [==============================] - 0s 237ms/step
1/1 [==============================] - 0s 231ms/step
1/1 [==============================] - 0s 223ms/step
1/1 [==============================] - 0s 229ms/step
1/1 [==============================] - 0s 236ms/step
1/1 [==============================] - 0s 239ms/step
1/1 [==============================] - 0s 254ms/step
1/1 [==============================] - 0s 241ms/step
1/1 [==============================] - 0s 221ms/step
1/1 [==============================] - 0s 213ms/step
1/1 [==============================] - 0s 236ms/step
1/1 [==============================] - 0s 226ms/step
1/1 [==============================] - 0s 233ms/step
1/1 [==============================] - 0s 242ms/step
1/1 [==============================] - 0s 242ms/step
1/1 [==============================] - 0s 241ms/step
1/1 [==============================] - 0s 227ms/step
```

```
1/1 [==============================] - 0s 281ms/step
1/1 [==============================] - 0s 318ms/step
1/1 [==============================] - 0s 260ms/step
1/1 [==============================] - 0s 276ms/step
1/1 [==============================] - 0s 259ms/step
1/1 [==============================] - 0s 238ms/step
1/1 [==============================] - 0s 233ms/step
1/1 [==============================] - 0s 244ms/step
1/1 [==============================] - 0s 225ms/step
1/1 [==============================] - 0s 244ms/step
1/1 [==============================] - 0s 247ms/step
1/1 [==============================] - 0s 240ms/step
1/1 [==============================] - 0s 204ms/step
1/1 [==============================] - 0s 216ms/step
1/1 [==============================] - 0s 224ms/step
1/1 [==============================] - 0s 234ms/step
1/1 [==============================] - 0s 231ms/step
1/1 [==============================] - 1s 628ms/step
1/1 [==============================] - 0s 260ms/step
1/1 [==============================] - 0s 235ms/step
1/1 [==============================] - 0s 254ms/step
1/1 [==============================] - 0s 241ms/step
1/1 [==============================] - 0s 239ms/step
1/1 [==============================] - 0s 246ms/step
1/1 [==============================] - 0s 241ms/step
1/1 [==============================] - 0s 253ms/step
1/1 [==============================] - 0s 257ms/step
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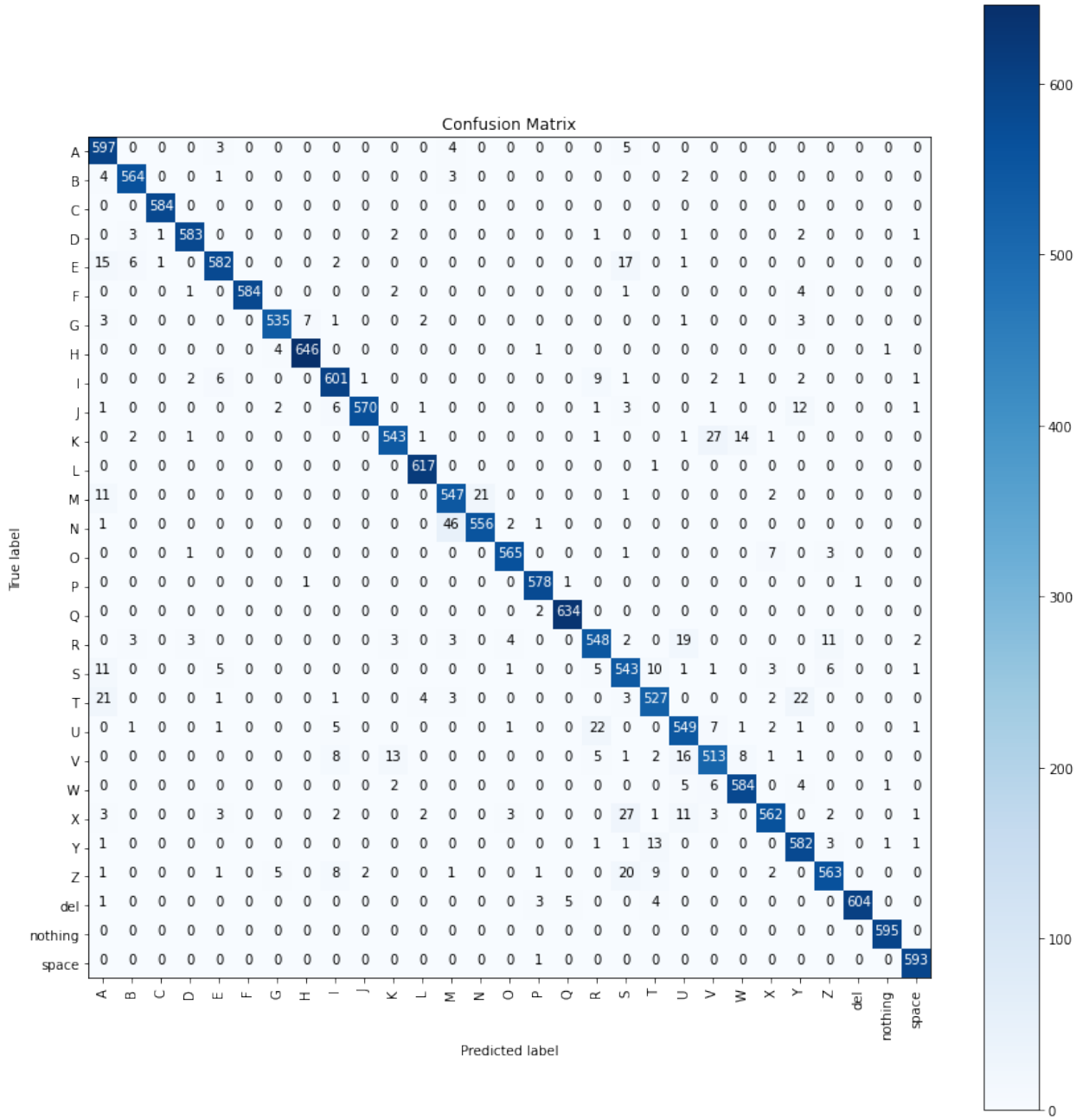
```python
# Plot the confusion matrix
def plot_confusion_matrix(cm, classes,
                          normalize=False,
                          title='Confusion matrix',
                          cmap=plt.cm.Blues):
    """
    This function prints and plots the confusion matrix.
    Normalization can be applied by setting `normalize=True`.
    """
    plt.figure(figsize = (12,12))
    plt.imshow(cm, interpolation='nearest', cmap=cmap)
    plt.title(title)
    plt.colorbar()
    tick_marks = np.arange(len(classes))
    plt.xticks(tick_marks, classes, rotation=90)
    plt.yticks(tick_marks, classes)
    if normalize:
        cm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]

    thresh = cm.max() / 2.
    for i, j in itertools.product(range(cm.shape[0]),
range(cm.shape[1])):
        plt.text(j, i, cm[i, j],
                 horizontalalignment="center",
                 color="white" if cm[i, j] > thresh else "black")
    plt.tight_layout()
    plt.ylabel('True label')
```

```
    plt.xlabel('Predicted label')

# Plot the confusion matrix
plt.figure(figsize=(10, 10))
plot_confusion_matrix(confusion_mtx, classes = class_names,
title='Confusion Matrix')
plt.show()
```

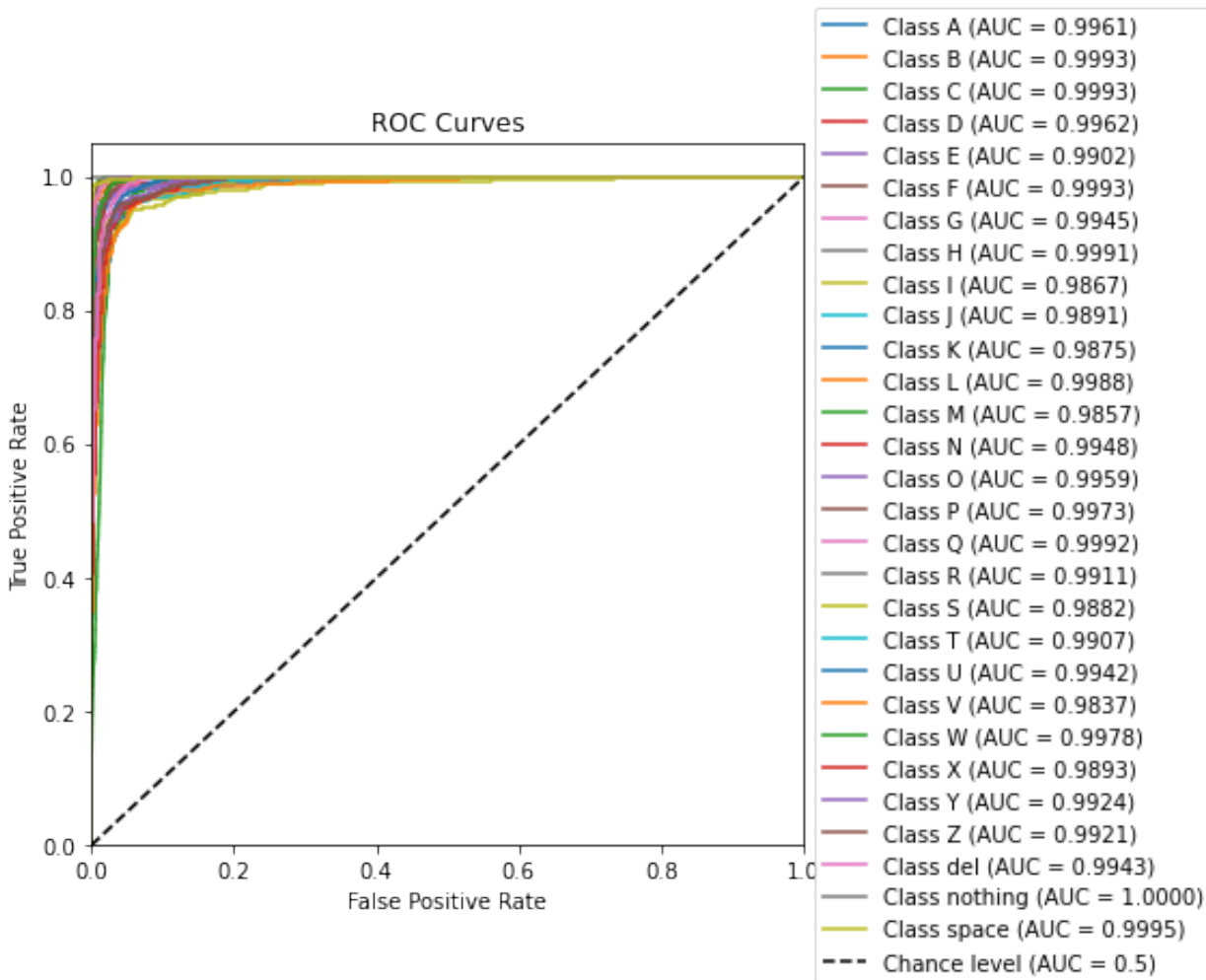
```
<Figure size 720x720 with 0 Axes>
```



```python
from sklearn.metrics import roc_auc_score, roc_curve
import matplotlib.pyplot as plt

# Calculate the AUC for each class
auc_scores = {}
for i in range(num_classes):
    auc_scores[class_names[i]] = roc_auc_score(y_true == i, y_pred[:,
i])

# Print the AUC scores for each class
for class_name, auc_score in auc_scores.items():
    print(f"AUC for class {class_name}: {auc_score:.4f}")

# Plot the ROC curves for each class
plt.figure(figsize=(8, 6))
for i in range(num_classes):
    fpr, tpr, _ = roc_curve(y_true == i, y_pred[:, i])
    plt.plot(fpr, tpr, label=f"Class {class_names[i]} (AUC =
{auc_scores[class_names[i]]:.4f})")

# Plot the chance level
plt.plot([0, 1], [0, 1], 'k--', label='Chance level (AUC = 0.5)')

plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curves')

# Adjust the plot to make space for the legend on the right
plt.subplots_adjust(right=0.7)

# Place the legend to the right of the plot
plt.legend(loc='center left', bbox_to_anchor=(1, 0.5))

plt.show()
```

```
AUC for class A: 0.9961
AUC for class B: 0.9993
AUC for class C: 0.9993
AUC for class D: 0.9962
AUC for class E: 0.9902
AUC for class F: 0.9993
AUC for class G: 0.9945
AUC for class H: 0.9991
AUC for class I: 0.9867
AUC for class J: 0.9891
AUC for class K: 0.9875
AUC for class L: 0.9988
AUC for class M: 0.9857
```

```
AUC for class N: 0.9948
AUC for class O: 0.9959
AUC for class P: 0.9973
AUC for class Q: 0.9992
AUC for class R: 0.9911
AUC for class S: 0.9882
AUC for class T: 0.9907
AUC for class U: 0.9942
AUC for class V: 0.9837
AUC for class W: 0.9978
AUC for class X: 0.9893
AUC for class Y: 0.9924
AUC for class Z: 0.9921
AUC for class del: 0.9943
AUC for class nothing: 1.0000
AUC for class space: 0.9995
```




```
model.evaluate(train_dataset)
```

```
/home/madeng/.local/lib/python3.10/site-packages/keras/src/
backend.py:5727: UserWarning: "`sparse_categorical_crossentropy`
received `from_logits=True`, but the `output` argument was produced by
a Softmax activation and thus does not represent logits. Was this
intended?
  output, from_logits = _get_logits(

1633/1633 [==============================] - 375s 229ms/step - loss:
0.3374 - accuracy: 0.9602

[0.33742958307266235, 0.9601576924324036]

model.evaluate(test_dataset)

543/543 [==============================] - 167s 237ms/step - loss:
0.3405 - accuracy: 0.9588

[0.340498149394989, 0.958832323551178]

model.evaluate(validation_dataset)

543/543 [==============================] - 169s 254ms/step - loss:
0.3414 - accuracy: 0.9601

[0.3414272964000702, 0.9600598812103271]
```